INFORME

REPORT

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1. FO-OEI-004 V03

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1. OBJECTIVES

- Assess the residual risk associated with potential part detachment in the GRIS robot during operation.
- Define the residual risk classification for each assessed item for generator inspection service.
- Document the assessment method, rating criteria, and acceptance basis used in the evaluation.

2. INTRODUCTION

Any detached or loosened part inside a generator represents a serious event, with the potential to damage internal components, reduce machine availability, and require recovery actions before return to service. Preventing this condition was therefore established as a primary design criterion throughout the development of the GRIS robot. Accordingly, the system was designed with specific retention, containment, and verification measures to minimize this risk. Nevertheless, as with any engineered system, a residual risk remains. This report assesses that residual risk for components that could detach during operation and formally documents the resulting evaluation.

3. ASSESSMENT METHOD

This assessment was performed as a residual risk evaluation of the released GRIS configuration. It assumes that the implemented retention, containment, and verification measures are already in place and effective under the standard service process, including application of the standard pre-test and post-test inspection checklist defined in document 00503-00-ZRQ001.

Each assessed item was evaluated independently, based on its specific detachment concern, implemented control measures, and potential consequence if the item were to remain inside the generator. The assessment does not treat the robot as a single generic assembly; instead, each item is considered as an individual case.

The numerical ratings were assigned by an expert committee with expertise in generator systems and robotics, using engineering judgment based on the reviewed design, implemented retention features, inspection criteria, and available validation evidence. The rating process considered the expected behavior of each item under normal operation.

The residual risk score is calculated as:

$$R = L \times M \times S$$

where:

L = likelihood of detachment

M = multiplicity factor, representing the number of identical instances

S = consequence severity, representing the potential damage if the machine were started with the assessed item still inside

This is a semi-quantitative engineering assessment, not a statistical failure-rate calculation. Its purpose is to provide a structured and technically justified comparison of the reviewed items and to determine whether the residual risk remains within the acceptable range for operation.

Acceptance is based on the resulting residual risk classification. Only items rated Very Low or Low are considered acceptable for the released configuration. Any higher classification requires additional review or mitigation.

4. ASSESSMENT SCALES

4.1. Likelihood

Value	Classification	Definition
0.1	Extremely Remote	Detachment is not considered credible under normal service conditions. It would require the failure of multiple independent barriers or an abrupt, unexpected break. Mechanical retention is maintained across all operating states.
0.5	Very Remote	Detachment is highly improbable under normal service conditions. It would require the failure of at least two independent retention features or a very localized loss of retention.
1.0	Remote	Detachment is remotely possible under normal service conditions. The item is fixed, non-adjustable between operations, and defined measures are in place to prevent loosening or release.
2.0	Unlikely	Detachment is improbable under normal service conditions. The item is manipulated during operation or service, but defined measures are in place to prevent loosening or release.
3.0	Possible	Detachment is possible under normal service conditions because the item lacks sufficient retention features or relies on a single control barrier.

4.2. Multiplicity factor

Value	Classification	Definition
1.0	Low Multiplicity	1–2 identical instances of the assessed item could be affected by the same failure mode.
2.0	Medium Multiplicity	3–5 identical instances of the assessed item could be affected by the same failure mode.
3.0	High Multiplicity	>5 identical instances of the assessed item could be affected by the same failure mode.

4.3. Consequence severity

Value	Classification	Definition
0.1	Negligible	No damage is expected. Recovery of the item is not required.
0.5	Minor	Visible marks or superficial damage may occur, but machine functionality is not expected to be affected. The item may remain in place, although removal should be considered.
1.0	Moderate	Minor damage may occur. The item must be removed.
2.0	Major	Significant damage is expected. The item must be removed.
3.0	Catastrophic	Catastrophic damage is expected. The item must be removed.

4.4. Residual risk classification

Range	Classification	Definition	Acceptability
0 < R ≤ 1.0	Very Low	No additional action required beyond implemented controls and standard verification.	Acceptable
1.0 < R ≤ 2.0	Low	Maintain implemented controls and apply focused attention during routine inspections.	Acceptable
2.0 < R ≤ 4.0	Moderate	Further justification or strengthened controls are required before release.	Not acceptable
4.0 < R ≤ 6.0	High	Additional mitigation is required before release.	Not acceptable
R > 6.0	Very High	Redesign or additional mandatory barriers are required before release.	Not acceptable

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5. ASSESSMENT RESULTS

5.1. Position Identification

The following figures identify the assessed positions on the GRIS robot used throughout this report and in the associated inspection checklist. Figure 1 shows the isometric top view, and Figure 2 shows the isometric bottom view.

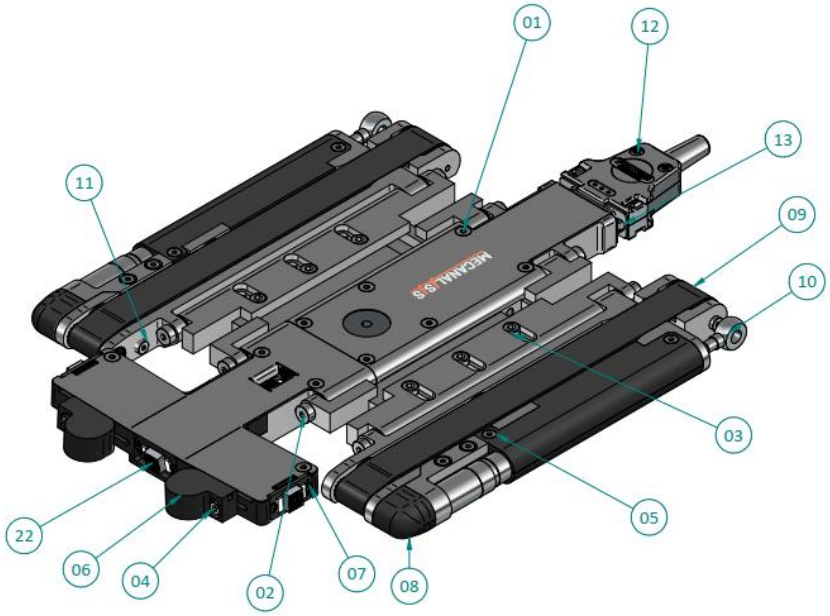


Figure 1. Position identification - GRIS isometric top view.

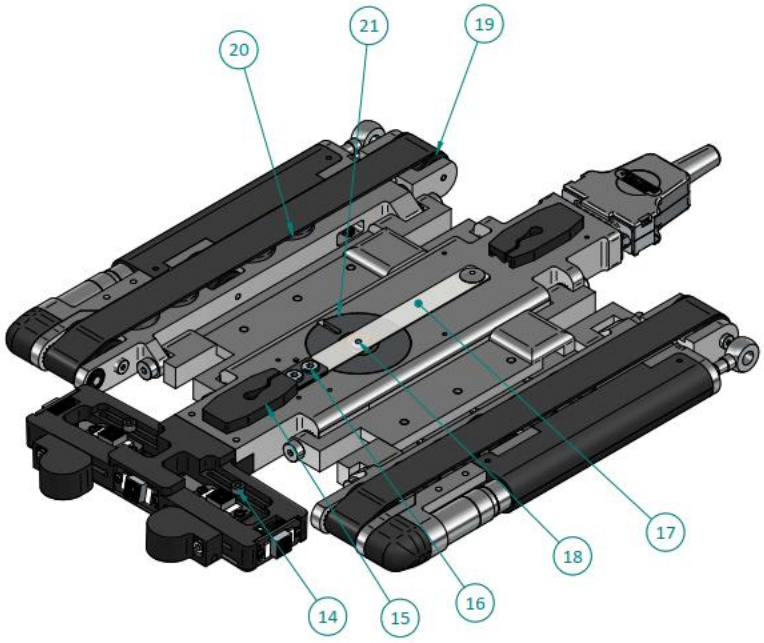


Figure 2. Position identification - GRIS isometric bottom view.

5.2. Residual Risk Assessment by Position

The following table presents the residual risk assessment results for each assessed position in the GRIS robot:

Pos.	Item	Detachment concern	Implemented controls	L	M	S	R	Class
1	Cover retaining screw	Backing out	Threadlocker + Torque mark + Retention tape over screw set	1.0	3	0.5	1.5	Low
2	Module joining shoulder screw	Backing out	Torque control	0.5	3	1.0	1.5	Low
3	Extender retaining screw	Backing out	Mechanical retention + Threadlocker + torque mark	0.1	1	1.0	0.1	Very Low
4	ELCID magnetic sensor fastening screw	Backing out / Pull-out	Mechanical retention	0.1	1	1.0	0.1	Very Low
5	Motor cover screw	Backing out / Pull-out	Threadlocker + torque mark + Retention tape over screw set	0.5	3	1.0	1.5	Low
6	ELCID magnetic sensor	Fracture	Material integrity inspection	0.5	1	1.0	0.5	Very Low
7	Camera retaining screw	Backing out / Pull-out	Threadlocker + Torque mark	1.0	3	0.5	1.5	Low
8	Gear cover	Fracture / Disengagement	Fit retention + Retention tape	1.0	1	0.1	0.1	Very Low
9	Track belt	Derailment	Validated track confinement; full derailment not credible	1.0	1	1.0	1.0	Very Low
10	Recovery hook	Backing out / Pull-out	Threadlocker + Torque mark	0.1	1	2.0	0.2	Very Low
11	Wheel axle shoulder screw	Backing out	Threadlocker + Torque mark	0.5	2	2.0	2.0	Low
12	Main connector housing screws	Backing out	Threadlocker + Torque mark + Retention tape over screw set	0.5	2	1.0	1.0	Very Low
13	Main connector latch screws	Backing out	Mechanical retention	0.1	1	1.0	0.1	Very Low
14	Bottom camera position locking screw	Backing out	Mechanical retention	0.1	1	1.0	0.1	Very Low
15	Central slot guide	Disengagement	Fit retention	2.0	1	0.5	1.0	Very Low
16	Leaf spring retaining screw	Backing out	Threadlocker + Torque mark	2.0	1	1.0	2.0	Low
17	Leaf spring	Fracture	Material integrity inspection	1.0	1	1.0	1.0	Very Low
18	Impactor grub screw	Backing out	Threadlocker + Retention tape	3.0	1	0.1	0.3	Very Low
19	Track wheels	Fracture	Material integrity inspection	0.1	2	0.5	0.1	Very Low
20	Magnets retaining screw	Backing out	Threadlocker + Torque mark	0.5	2	2.0	2.0	Low
21	Impact solenoid	Detachment	Mechanical retention + Interference fit	0.1	1	3.0	0.3	Very Low
22	Camera lens	Backing out	Adhesive bond	3.0	1	0.1	0.3	Very Low

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6. CONCLUSION

The assessed GRIS configuration remains within the acceptable residual risk range for generator inspection service. All reviewed items were classified as either Very Low or Low, and no item reached a Moderate or higher risk level. The highest residual risk score obtained was 2.0, which remains within the acceptable range.

Based on this assessment, no additional mitigation is required for the released configuration prior to service. This conclusion remains valid only for the reviewed GRIS configuration, its implemented controls, and the current standard verification process. Any relevant design, retention, operating, or verification change shall require reassessment.

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